

■ Dart Star Pattern Programs

1. Simple Triangle

Code:

```
void main() {  
    int n = 5;  
    for (int i = 1; i <= n; i++) {  
        print('*' * i);  
    }  
}
```

Output:

```
*  
**  
***  
****  
*****
```

2. Right-Aligned Triangle

Code:

```
void main() {  
    int n = 5;  
    for (int i = 1; i <= n; i++) {  
        print(' ' * (n - i) + '*' * i);  
    }  
}
```

Output:

```
    *  
   **  
  ***  
 ****  
*****
```

3. Pyramid

Code:

```
void main() {  
    int n = 5;  
    for (int i = 1; i <= n; i++) {  
        print(' ' * (n - i) + '*' * (2 * i - 1));  
    }  
}
```

Output:

```
    *  
   ***  
  *****  
 *****  
*****
```

4. Inverted Triangle

Code:

```

void main() {
    int n = 5;
    for (int i = n; i >= 1; i--) {
        print('*' * i);
    }
}

```

Output:

```

*****
****
***
**
*

```

5. Diamond

Code:

```

void main() {
    int n = 5;
    for (int i = 1; i <= n; i++) {
        print(' ' * (n - i) + '*' * (2 * i - 1));
    }
    for (int i = n - 1; i >= 1; i--) {
        print(' ' * (n - i) + '*' * (2 * i - 1));
    }
}

```

Output:

```

    *
   ***
  *****
 *****
*****
*****
 *****
  *****
   ***
    *

```

6. Square

Code:

```

void main() {
    int n = 5;
    for (int i = 1; i <= n; i++) {
        print('* ' * n);
    }
}

```

Output:

```

* * * * *
* * * * *
* * * * *
* * * * *
* * * * *

```

7. Hollow Square

Code:

```

void main() {
    int n = 5;
    for (int i = 1; i <= n; i++) {
        if (i == 1 || i == n) {

```

```

        print('* ' * n);
    } else {
        print('* ' + ' ' * (n - 2) + '*');
    }
}
}

```

Output:

```

* * * * *
*         *
*         *
*         *
*         *
* * * * *

```

8. Right-Angled Hollow Triangle

Code:

```

void main() {
    int n = 5;
    for (int i = 1; i <= n; i++) {
        if (i == 1 || i == n) {
            print('*' * i);
        } else {
            print('*' + ' ' * (i - 2) + '*');
        }
    }
}

```

Output:

```

*
**
* *
*  *
*****

```

9. Inverted Pyramid

Code:

```

void main() {
    int n = 5;
    for (int i = n; i >= 1; i--) {
        print(' ' * (n - i) + '*' * (2 * i - 1));
    }
}

```

Output:

```

*****
*****
*****
***
*

```

10. Hollow Pyramid

Code:

```

void main() {
    int n = 5;
    for (int i = 1; i <= n; i++) {
        if (i == n) {
            print('*' * (2 * i - 1));
        } else {
            print(' ' * (n - i) + '*' + ' ' * (2 * i - 3 > 0 ? 2 * i - 3 : 0) + (i > 1 ? '*' : ''));
        }
    }
}

```

```
    }
  }
}
```

Output:

```

      *
     * *
    *   *
   *     *
  *       *
 *         *
*****
```

11. Sandglass Pattern

Code:

```
void main() {
    int n = 5;
    for (int i = n; i >= 1; i--) {
        print(' ' * (n - i) + '*' * (2 * i - 1));
    }
    for (int i = 2; i <= n; i++) {
        print(' ' * (n - i) + '*' * (2 * i - 1));
    }
}
```

Output:

```

*****
*****
*****
***
*
***
*****
*****
*****
```

12. Hourglass

Code:

```
void main() {
    int n = 5;
    for (int i = n; i >= 1; i--) {
        print('*' * i);
    }
    for (int i = 2; i <= n; i++) {
        print('*' * i);
    }
}
```

Output:

```

*****
****
***
**
*
**
***
****
*****
```

13. Hollow Diamond

Code:

```

void main() {
    int n = 5;
    for (int i = 1; i <= n; i++) {
        print(' ' * (n - i) + '*' + ' ' * (2 * i - 3 > 0 ? 2 * i - 3 : 0) + (i > 1 ? '*' : ''));
    }
    for (int i = n - 1; i >= 1; i--) {
        print(' ' * (n - i) + '*' + ' ' * (2 * i - 3 > 0 ? 2 * i - 3 : 0) + (i > 1 ? '*' : ''));
    }
}

```

Output:

```

    *
   * *
  *   *
 *     *
*       *
 *     *
  *   *
   * *
    *

```

14. Butterfly Pattern

Code:

```

void main() {
    int n = 5;
    for (int i = 1; i <= n; i++) {
        print('*' * i + ' ' * (2 * (n - i)) + '*' * i);
    }
    for (int i = n; i >= 1; i--) {
        print('*' * i + ' ' * (2 * (n - i)) + '*' * i);
    }
}

```

Output:

```

*           *
**          **
***         ***
****        ****
*****       *****
*****       *****
****        ****
***         ***
**          **
*           *

```